Rishab Premod Nambiar

rishabpnambiar@google.com + 91 8848862626 Rishab P Nambiar RPNambiar-24 Portfolio

Machine Learning-focused Data Science student building end-to-end AI systems across NLP, Computer Vision and Reinforcement Learning. Experienced in Transformers, GANs, XGBoost, and Explainable AI, with projects involving model deployment and real-world datasets. Seeking an ML/DS internship to develop and deploy data-driven systems.

Education

Amrita Vishwa Vidyapeetham, Coimbatore *Sept 2022 – Present*
4th Year Integrated MSc Data Science

- **GPA :** 8.99/10
- **Coursework:** Deep Learning, Machine Learning, Data Visualization, Statistics, Database Management System, Data Wrangling, Image Processing

Kendriya Vidyalaya No 1, Calicut *2020-21*
AISSCE

- **Marks :** 95.2
- **Subjects :** Physics, Chemistry, Maths, Biology

Machine Learning Projects

Image Colorization using GANs

- Built an end-to-end image colorization pipeline using a Pix2Pix GAN (Generator–Discriminator) to convert grayscale images into realistic color outputs.
- Improved visual fidelity and color consistency through optimized adversarial training, achieving PSNR: 24.56 and SSIM: 0.899.
- Designed preprocessing and evaluation workflows in Python to enable reproducible experiments and metric-based model comparison.

Adaptive Traffic Signal Controller using Reinforcement Learning

- Developed a Deep Q-Network (DQN) based traffic signal controller for a simulated 2-way intersection, learning optimal phase-switching policies from environment rewards.
- Reduced congestion ($\sim$10%) vs fixed-time baseline and improved intersection throughput.
- Built the simulation environment and real-time visualization using PyGame, enabling iterative training and policy evaluation.

DetectAI: Human vs AI-Generated Text Classifier

- Engineered a hybrid AI-text detection system combining TF-IDF + Random Forest + BiLSTM with a fine-tuned RoBERTa transformer, achieving 98.64% accuracy.
- Integrated SHAP and LIME to provide word-level interpretability and improve trust in model predictions.
- Deployed an interactive web demo using Gradio, enabling real-time inference and user-facing testing.

MedXAI-CXR: Explainable Deep Learning for Chest X-ray Diagnosis

- Developed a multi-class chest X-ray classification system (COVID-19, Pneumonia, Normal) using a baseline CNN and transfer learning with ResNet50, achieving 97% accuracy.
- Integrated Explainable AI techniques (Grad-CAM, LIME, SHAP) to visualize model decision regions and evaluate clinical interpretability of predictions.
- Performed comparative analysis between shallow CNN and pretrained deep networks, demonstrating improved convergence, diagnostic sensitivity, and trustworthy model behavior.

Predictive Demand Modeling in the Logistics Industry

- Built a time-series forecasting pipeline to predict grocery sales using engineered temporal features and

optimized regression workflows.

- Trained and tuned XGBoost models to improve forecast stability and accuracy across varying demand patterns.
- Won SMILe Logistics Innovation Hack by delivering actionable forecasting insights and model-driven decision support.

Illegal Fence Detection System using AI for Wildlife Conservation

- Developed an AI-powered drone-based system to detect illegal electric fences in wildlife regions using YOLOv4 object detection.
- Improved detection reliability under field-like conditions through data preprocessing and model refinement.
- Presented as a paper at the Fifth International Conference on Artificial Intelligence & Smart Computing (ICAISC-2025)

Ongoing Projects

MockViewAI

- Building an AI-driven mock interview system that generates role-specific questions from resumes and job descriptions and evaluates responses.
- Implements automated scoring to evaluate candidate responses and provide structured feedback.

Malware Classifier using Zero Shot Learning

- Developing a robust Zero-Shot Learning malware classifier using MaleVis and MalImg datasets.
- Evaluating adversarial robustness using PGD / FGSM attacks, analyzing performance drops under perturbations.

Skills

- **ML / DL & AI:** Scikit-learn, XGBoost, PyTorch, Transformers, NLP, Computer Vision, Explainable AI (XAI)
- **Programming:** Python, R, SQL, MATLAB
- **Data Analysis:** NumPy, Pandas, SciPy
- **Visualization:** Matplotlib, Plotly, Tableau
- **Deployment:** Streamlit, Gradio

Additional Certifications

Python Certification - *Issued by HackerRank*

Problem Solving Certification - *Issued by HackerRank*

Extra Curricular Activities

- Part of the technical team that conducted Ideathon '24 - a Hackathon conducted during the Anokha Tech Fest of Amrita Vishwa Vidyapeetham, Ettimadai Campus
- Won fifth prize in group song (Indian) at the 38th AIU Inter University Youth Festival South Zone.
- Won fourth prize in group song (Indian) at the 39th AIU Inter University Youth Festival South Zone.